

PointNet meets Self-Attention Graph Pooling: A Synergistic Approach to Point Cloud Classification

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Abstract—The point cloud is a crucial 3D data structure that plays a significant role in environmental research and has extensive applications in industries such as autonomous driving and image segmentation. Additionally, graphs are useful data structures for regularizing point cloud data, allowing the incorporation of graph features to extract specific attributes for each point in the dataset. Graph-based neural networks facilitate the extraction of features from point cloud data, ensuring that each point's unique characteristics contribute to the classification results, thereby enhancing detection accuracy. In this paper, we investigate the impact of graph centrality features on each point using Point-Net and Self-Attention Graph Pooling as baselines. We also introduce two novel architectures for point cloud classification. The first architecture integrates the Point-Net and Self-Attention Graph Pooling models using fusion approaches. The second model is a Point-Net based Self-Attention Graph Pooling, which utilizes the results from the first model as input for the second. Extensive experiments conducted on the ModelNet10 dataset demonstrate that the inclusion of Closeness and Harmonic centralities are the best centralities for classification. Our new architectures improve classification accuracy by 3.5% compared to the individual architectures, while also reducing the model size by 1MB.

Index Terms—Point Cloud Classification, PointNet, Self-Attention Graph Pooling, 3D Object Recognition.

I. Introduction

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II. Related Works

[should be done by Ms.Atybai]

III. PROPOSED APPROACHES

A. Feature Engineering

In this section, we present our approach to feature engineering in point cloud classification due to the limitations of XYZ features. To enhance feature representation, we exploit graph-based features, starting with the construction of a weighted and directed graph using the k-nearest neighbors method. As we consider the inherent

3D spatial relationships, we set K to 6, representing the six absolute sides in 3D space. This graph enables efficient feature aggregation and captures crucial spatial dependencies among points.

To further enrich the feature set, we introduce node centrality as an essential feature for each node (point in space) within the graph. We employ seven centrality metrics, namely betweenness, closeness, Katz, PageRank, eigenvector, harmonic, and load centrality.

- **Betweenness:** This centrality feature measures the importance of a node in connecting other nodes. Mathematically, let $n_{s,t}^i$ be the number of geodesic paths from s to t that pass through i and let $n_{s,t}$ be the total number of geodesic paths from s to t . Recall that a geodesic path is not necessarily unique and the geodesic paths between a pair of vertices need not be node-independent, meaning they may pass through some of the same vertices. Then the betweenness centrality of vertex i is calculated by equation 1:

$$b_i = \sum_{s,t} w_{s,t}^i = \sum_{s,t} \frac{n_{s,t}^i}{n_{s,t}} \quad (1)$$

- **Closeness:** closeness centrality (or closeness) of a node is a measure of centrality in a network, calculated as the sum of the length of the shortest paths between the node and all other nodes in the graph. Thus the more central a node is, the closer it is to all other nodes. The closeness centrality of node x is calculated by equation 2. where $d(y, x)$ is the distance between vertices x and y .

$$C(x) = \frac{N}{\sum_y d(y, x)} \quad (2)$$

- **Katz:** Katz centrality computes the relative influence of a node within a network by measuring the number

of the immediate neighbors (first degree nodes) and also all other nodes in the network that connect to the node under consideration through these immediate neighbors. In summary it combines both local and global node information. For calculate Katz for a node Let $A = (a_{i,j})$ be the adjacency matrix of a directed graph. The Katz centrality x_i of node i is given by:

$$x_i = \alpha \sum_k a_{k,i} x_k + \beta$$

where α and β are constants. In matrix form we have:

$$x = \alpha x A + \beta$$

where β is now a vector whose elements are all equal a given positive constant. The free parameter α , known as the damping factor, should be chosen with case. For small (close to 0) values of α the contribution given by paths longer than one rapidly declines, and thus Katz scores are mainly influenced by short paths (mostly in-degrees) and by the exogenous factor of the system.

- **PageRank:** PageRank is an adjustment of Katz centrality that takes into consideration this issue. There are three distinct factors that determine the PageRank of a node: (i) the number of links it receives, (ii) the link propensity of the linkers, and (iii) the centrality of the linkers. The first factor is not surprising: the more links a node attracts, the more important it is perceived. For calculate PageRank for a node Let $A = (a_{i,j})$ be the adjacency matrix of a directed graph. The PageRank centrality x_i of node i is given by:

$$x_i = \alpha \sum_k \frac{a_{k,i}}{d_k} x_k + \beta$$

where α and β are constants and d_k is the out-degree of node k if such degree is positive, or $d_k = 1$ if the out-degree of k is null. In matrix form we have:

$$x = \alpha x D^{-1} A + \beta$$

where β is now a vector whose elements are all equal a given positive constant and D^{-1} is a diagonal matrix with i -th diagonal element equal to $1/d_i$. Notice that, as seen for Katz centrality, PageRank is determined by an endogenous component that takes into consideration the network topology and by an exogenous component that is independent of the network structure. It follows that x can be computed as:

$$x = \beta(I - \alpha D^{-1} A)^{-1}$$

The damping factor α and the personalization vector β have the same role seen for Katz centrality. In particular, α should be chosen between 0 and $1/\rho(D^{-1}A)$. Also, when the network is large, it is preferable to use the power method for the computation of PageRank.

- **Eigenvector:** Eigenvector centrality (also called eigencentrality) is a measure of the influence of a node in a network. It assigns relative scores to all nodes in the network based on the concept that connections to high-scoring nodes contribute more to the score of the node in question than equal connections to low-scoring nodes. PageRank and the Katz centrality are variants of the eigenvector centrality. Using the adjacency matrix to find eigenvector centrality For a given graph $G := (V, E)$ with $|V|$ vertices let $A = (a_{v,t})$ be the adjacency matrix, i.e. $a_{v,t} = 1$ if vertex v is linked to vertex t , and $a_{v,t} = 0$ otherwise. The relative centrality score of vertex v can be defined as:

$$x_v = \frac{1}{\lambda} \sum_{t \in M(v)} x_t = \frac{1}{\lambda} \sum_{t \in G} a_{v,t} x_t$$

- **Harmonic:** Harmonic centrality measures the 'average' distance of a node to the other nodes in the network. This can be used to find nodes in a network that can quickly communicate with other nodes. For calculation harmonic centrality for node i let $d(i, x)$ is a distance from node i to node x for a given graph $G := (V, E)$ with $|V|$ vertices then :

$$Harmonic(i) = \sum_{x \in G, x \neq i} \frac{1}{d(i, x)}$$

- **Load centrality:** The load centrality of a node is the fraction of all shortest paths that pass through that node. It measures the node's importance in graph connectivity and load distribution. For calculate load centrality Consider a graph $G := (V, E)$ and an algorithm to identify the (potentially multiple) minimum weight path(s) between any pair of vertices (s, d) . Let $\theta_{s,d}$ be a quantity of a commodity that is sent from vertex s to vertex d . We assume the commodity is always passed to the next hop following the minimum weight paths. In case of multiple next hops, the commodity is divided equally among them. We call $\theta_{(s,d)}(v)$ the amount of commodity forwarded by vertex v . The load centrality of v is then given by:

$$LC(v) = \sum_{s,d \in V} \theta_{(s,d)}(v)$$

In our feature engineering process, we combine the seven centrality features with the existing XYZ features (x, y, and z coordinates) for each node. This results in a feature vector comprising 10 distinctive elements per point in the point cloud. Additionally, we applied a 1024 sampling rate to each point cloud to handle varying point densities efficiently. To ensure uniformity, we performed normalization across the feature vectors. Furthermore, random noise was introduced to augment the dataset, enhancing the model's robustness to real-world scenarios and improving generalization performance. These feature engineering techniques aim to provide a comprehensive

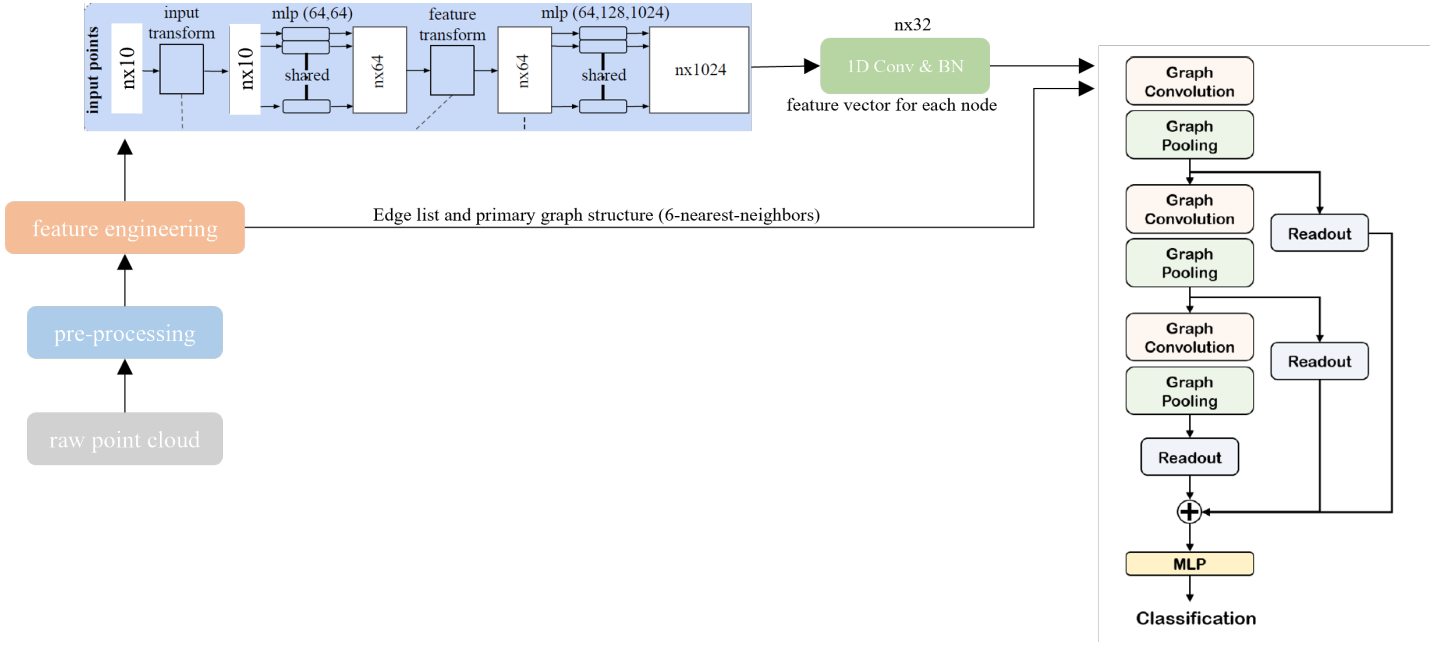


Fig. 1. The architecture for Feature Concatenation Approach.

and informative representation of the point cloud data, optimizing the classification performance in the subsequent stages of our research. As shown in the subsequent section, the features closeness and harmonic are identified as the most effective additions to the point cloud’s existing x , y , and z dimensions. These features were subsequently utilized to advance our approach.

It is important to note that the calculation of feature vectors for node centralities was implemented using multiprocessing on the CPU. This approach allowed us to efficiently compute centrality metrics for each node in the point cloud concurrently, significantly reducing the computation time. By leveraging the power of multiprocessing, we were able to accelerate the feature engineering process and efficiently handle large datasets, ultimately improving the overall efficiency and performance of the fusion approaches.

We selected Point-Net and Self-Attention Graph Pooling models as baselines for improvement. Point-Net is a general benchmark model for point cloud classification, suggesting that any enhancements achieved with this baseline could potentially be replicated with other models. Similarly, Self-Attention Graph Pooling is a popular benchmark in graph classification, and achieving better results with this model could indicate improved generalization across other baselines.

B. Feature Concatenation Approach

In this proposed fusion approach, named “FeatureConcatModel,” we combine the high-level features obtained from PointNet and SAGPool networks for efficient point cloud classification. Both networks take point clouds with

10 defined features as input. SAGPool outputs vectors with 64 features for each point cloud in a batch.

To adapt PointNet, we introduce a linear layer, reducing the 1024 features to 64 features for each point cloud. Subsequently, we concatenate the 64-feature vectors from both PointNet and SAGPool, creating a unified vector with 128 features. This concatenated feature vector captures both the local and global information, ensuring a comprehensive representation of the point cloud data. To further process the concatenated features, we employ three multi-layer perceptron (MLP) networks. The first MLP contains 64 neurons, followed by a second MLP with 32 neurons, and finally a third MLP with 10 neurons corresponding to the number of classes. Batch normalization is applied to all three MLPs, enhancing the model’s stability and accelerating convergence.

We employ dropout techniques with a probability of 0.3 in the second layer, reducing overfitting and improving generalization performance. Additionally, we incorporate the Graph Attention (GAT) layer from SAGPool with 6 attention heads, considering the 6 nearest neighbors for each point.

This proposed “FeatureConcatModel” fusion approach offers a promising solution for point cloud classification, leveraging the strengths of both PointNet and SAGPool architectures. By effectively combining their high-level features, we aim to achieve improved classification accuracy and robustness in handling complex point cloud data. The architecture for feature concatenation is presented in “Fig. 1”.

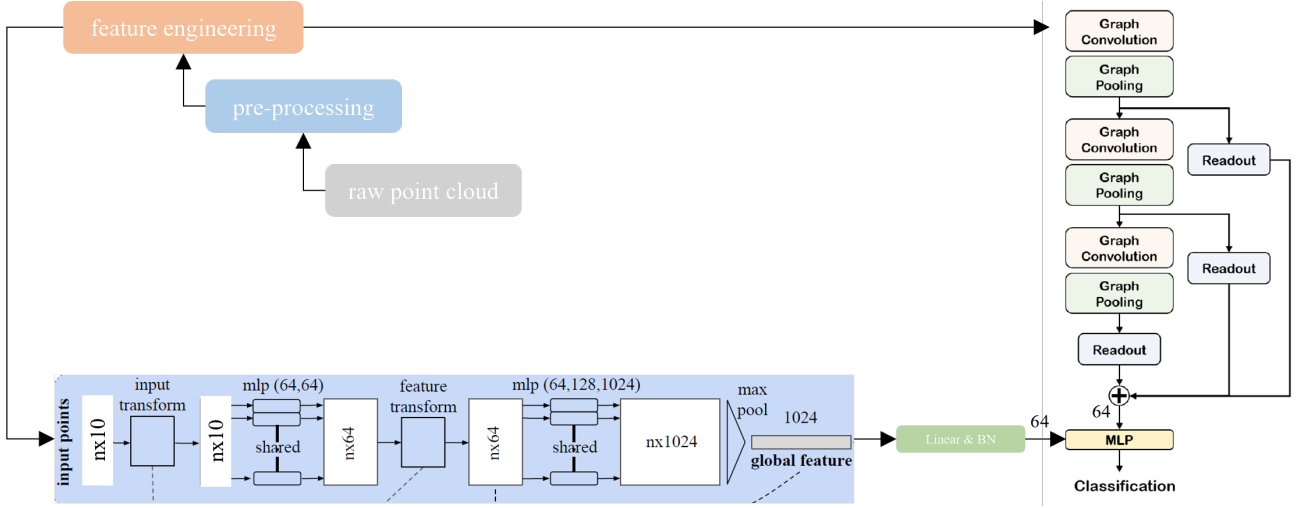


Fig. 2. The architecture for the graph classification based on PointNet features.

C. Graph Classification based on PointNet Features

The "PointNetBasedGraphPoolingModel" fusion approach represents a sophisticated integration of SAGPool and PointNet architectures, harnessing the strengths of both networks to enhance point cloud classification. To begin, we utilize a set of 10 defined features, including XYZ coordinates and centrality metrics, as input to the custom version of PointNet. In this custom implementation, we intentionally omit the global pooling section to obtain a 1024-feature vector for each node in the point cloud. This design choice enables PointNet to capture intricate local features and spatial relationships present in the data. The architecture for the graph classification based on PointNet features is showcased in "Fig. 2".

Subsequently, we enhance the feature vectors obtained from PointNet by applying 1D convolution, followed by batch normalization. This operation efficiently reduces the dimensionality to 32, effectively distilling essential information from PointNet's output. The resulting 32-feature vectors, carrying valuable local feature representations, are then fed into the SAGPool network.

SAGPool, with its primary edge list structure and consistent graph schema, efficiently performs adaptive feature pooling and aggregation. The integration of the PointNet-derived feature vectors into the SAGPool network further enhances its ability to capture meaningful global context while preserving crucial local details.

Experimental evaluations of the "PointNetBasedGraphPoolingModel" approach showcase its impressive performance in point cloud classification. By seamlessly blending the local feature extraction capabilities of PointNet with the adaptive feature aggregation of SAGPool, the approach achieves a more comprehensive representation of the point cloud data. The fusion of these two architectures leads to notable improvements in classification accuracy, enabling more robust, reducing the model size and accu-

rate recognition of complex 3D scenes and objects.

The effectiveness of this fusion approach contributes significantly to the advancement of 3D perception and scene understanding applications. With its ability to leverage both global and local features, the "PointNet-BasedGraphPoolingModel" approach exemplifies a powerful paradigm in point cloud classification and opens new avenues for future research and practical applications in various domains.

IV. EXPERIMENTS AND RESULTS

In this section, we provide a detailed account of the training process for the four networks aimed at investigating the performance of the intersection of SAGPool and PointNet compared to each architecture used individually. For the first network, a traditional PointNet was trained using the standard ModelNet10 dataset with 3 features (XYZ coordinates). The results of this training are presented in "Fig. 3".

A. Dataset

The dataset utilized for training and testing the networks in this study is ModelNet, a widely used benchmark in the field of 3D object recognition and classification. ModelNet comprises ModelNet10 and ModelNet40, which feature 10 and 40 classes, respectively. These datasets include categories of common objects such as chairs, tables, airplanes, and cars, among others. To prepare the dataset for training and testing, several preprocessing steps were applied. Firstly, we employed a 1024 sampling rate to obtain a fixed number of points from each 3D model. This step ensured a consistent input size for the networks, facilitating efficient processing. Next, we applied normalization to scale the point coordinates, making them relative to the model's bounding box and ensuring that they lie within a common range. This normalization step

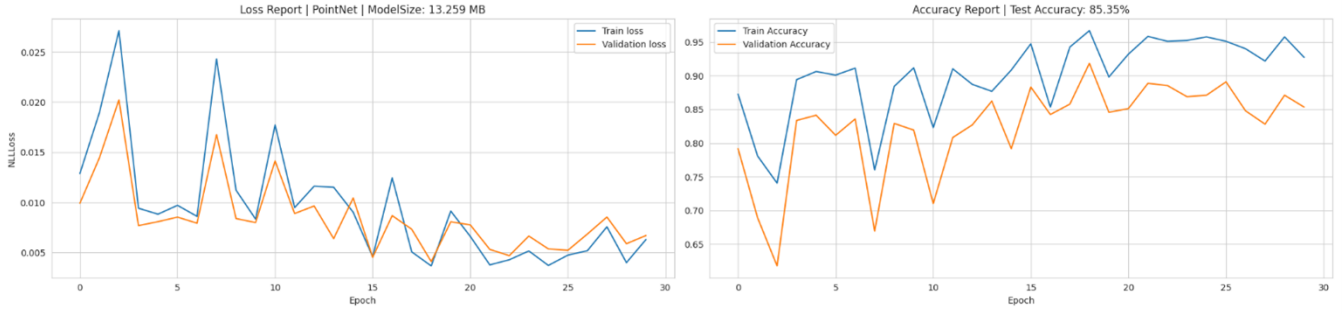


Fig. 3. Training result for traditional PointNet Network.

enhanced the model’s robustness and improved convergence during training.

Additionally, to augment the dataset and enhance the model’s generalization capabilities, we introduced random noise to the point cloud data. This noise addition introduced small variations in the point positions, mimicking real-world scenarios and improving the network’s ability to handle noisy input data. Furthermore, random rotation was applied to the 3D models during training. This augmentation technique involved rotating the objects along different axes, creating diverse orientations for each model. By doing so, the network learned to be invariant to object rotations, improving its ability to recognize and classify objects from different viewpoints.

To ensure an unbiased evaluation of the models’ performance, the original ModelNet datasets was divided into predefined training and testing sets. The training set was used to train the fusion approaches, while the testing set was kept separate for evaluation. This separation allowed for an objective assessment of the models’ classification accuracy and generalization to unseen data. The utilization of the ModelNet10 dataset, combined with the various preprocessing and augmentation techniques, contributed to the robustness and effectiveness of the trained networks in point cloud classification tasks.

B. Training

The second network involved a hierarchical version of SAGPool with a custom improvement that incorporated learning weights for each stage. We trained this network using a feature vector of 10 dimensions, which included the XYZ coordinates along with centrality metrics. The training outcomes for this hierarchical SAGPool network can be observed in “Fig. 4”.

The third network implemented our proposed feature concatenation approach, where we combined the high-level features extracted from both PointNet and SAGPool networks.

Similarly, a feature vector of 10 dimensions, encompassing XYZ coordinates and centrality metrics, was used for training. The training results for this feature concatenation approach are depicted in Fig. 5”. Lastly, we explored the graph classification based on PointNet features approach as our fourth network. Again, a feature vector of 10 dimensions, comprising XYZ coordinates and centrality metrics, was utilized for training. The training outcomes for this graph-based classification approach are illustrated in Fig. 6”. In all four networks, the same sets were used for training and testing, ensuring a fair and unbiased evaluation of their performance. The inclusion of centrality metrics in the feature vectors enhances the models’ capability to capture essential node connectivity information, contributing to improved classification accuracy. These comprehensive experiments allow us to gain

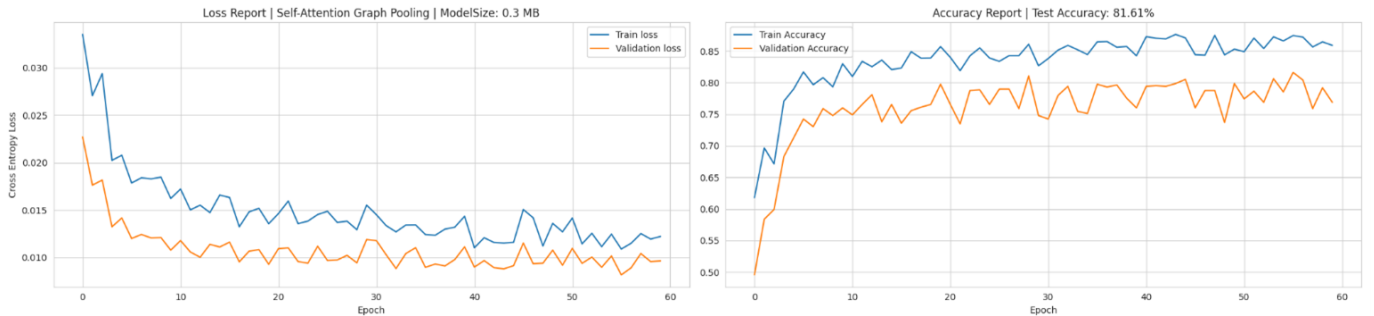


Fig. 4. Training result for original hierarchical SAGPool network.

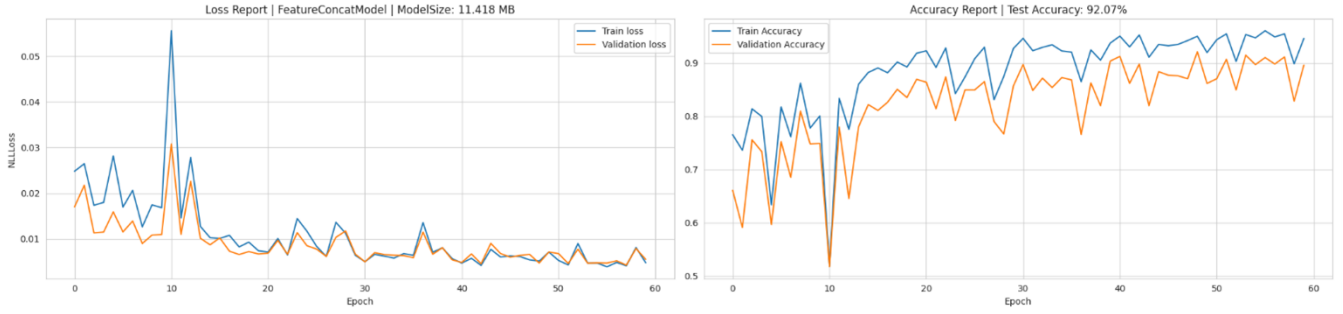


Fig. 5. Training result for feature concatenation approach.

valuable insights into the efficacy of combining SAGPool and PointNet architectures and highlight their potential to enhance point cloud classification tasks. The results obtained from these experiments play a pivotal role in shaping our understanding of the proposed fusion approaches and their practicality in real-world applications. Also the summary of the training results is presented in Table. I

C. Implementation details and hyper-parameters

In our implementation, we adopted the PyTorch framework as the primary tool for model development. To facilitate graph pooling and centrality calculations, we incorporated PyTorch Geometric and NetworkX libraries, respectively. A comprehensive dataloader was designed, encompassing all

essential details, including XYZ features, graph edge lists, centrality features, and labels.

To optimize processing costs and expedite training, we employed parallel computation to precalculate and store the required features in separate files. During training, we efficiently loaded the precomputed features from the drive, minimizing redundant calculations and streamlining the training process.

For training and testing, we set the batch size to 32 and 64, respectively. Across all networks, a graph pooling rate of 0.25 was consistently applied, along with a Graph Attention (GAT) convolution layer featuring 6 heads and 128 hidden features.

The ADAM optimizer was chosen for training, with weight decay 0.0005 and learning rates of 0.01, 0.001, and 0.005 for the SAGPool, PointNet, and fusion approaches, respectively. Due to resource limitations during the research, we trained the networks for 30-60 epochs, allowing us to explore a wide range of model performance within our constraints.

Our experiments were conducted on a Desktop PC with a robust hardware configuration. The machine featured an Intel Core i7 12700K CPU, operating at 4.9 GHz frequency, and 32GB DDR5 main memory. Additionally, an Nvidia 3070TI GPU with 8GB memory was utilized for accelerated computations. Notably, having access to a GPU with higher memory would have facilitated training with larger batch sizes and enabled longer model training, potentially leading to even more promising results.

In summary, our implementation details and well-optimized hyperparameters ensured efficient and effective exploration of the fusion approaches' performance. While resource constraints influenced certain aspects of our experiments, the chosen hardware configuration allowed us to achieve substantial progress in understanding the potential benefits of combining SAGPool and PointNet architectures for point cloud classification tasks.

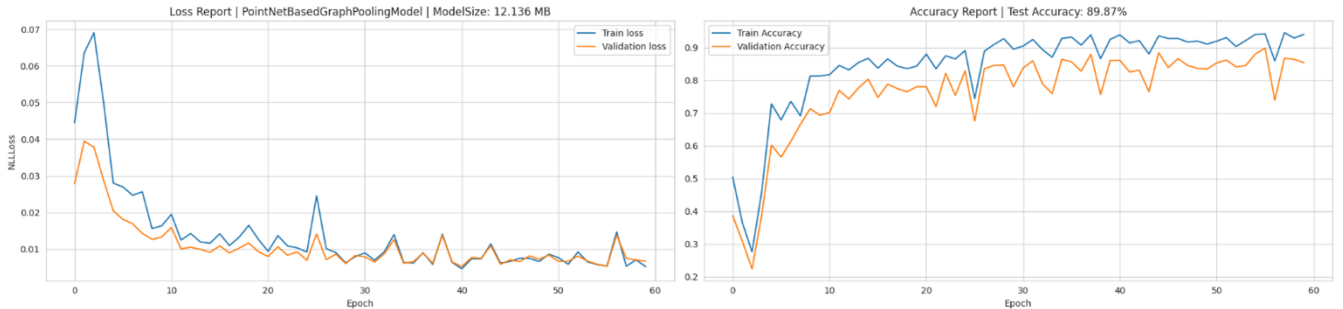


Fig. 6. Training result for graph classification based on PointNet features approach.

TABLE I

Training Results of Four Networks. This table provides a comprehensive overview of the results obtained from training four networks, including traditional PointNet, hierarchical SAGPool, feature concatenation, and Graph Classification based on PointNet Features approaches. For each network, the features used during training, test accuracy achieved, model size, and corresponding learning rate are reported. The best-performing results are denoted by bold font, while the second-best results are underlined, offering valuable insights into the performance of each approach

Network	Features	Acc	Size	LR
PointNet	3 (XYZ)	85.3	13.259 MB	0.001
Hierarchical SAGPool	3 (XYZ) + 7 (Centralities)	81.61	0.3 MB	0.01
FeatureConcat	3 (XYZ) + 7 (Centralities)	92.07	11.418 MB	0.005
PointNetBasedGraphPooling	3 (XYZ) + 7 (Centralities)	89.87	12.136 MB	0.005

V. CONCLUSION

A. Future works

- **Diverse Dataset Exploration:** To further validate the robustness and generalization of the proposed fusion approaches, it is essential to explore and evaluate their performance on different datasets with varying complexities and sizes. Investigating datasets that encompass a wider range of 3D objects and scenes will provide deeper insights into the approaches' adaptability and effectiveness in real-world scenarios.
- **Extended Training Analysis:** In future research, the effect of extended training periods should be explored to understand how longer training durations impact the convergence and overall performance of the fusion approaches. Additionally, investigating the influence of different batch sizes and learning rates on training dynamics and model performance will help in optimizing the training process and achieving higher accuracy.
- **Fine-tuning of Fusion Approach Components:** The proposed fusion approaches, FeatureConcat and PointNetBasedGraphPooling, utilize MLP and 1D convolution layers, dropout, and batch normalization. Fine-tuning these components and hyperparameters can further enhance the network's performance and efficiency. Optimizing these details will lead to improved feature extraction and pooling, contributing to better classification results. The source code for the proposed fusion approaches, along with the experimental setup and dataset preparation, is available at the following link: <https://github.com/MohsenEbadpour/PointNet-meets-Self-Attention-Graph-Pooling-A-Synergistic-Approach-to-Point-Cloud-Classification>. Researchers are encouraged to contribute to the development and completion of this work. Collaborating on refining the fusion approaches, exploring additional datasets, and conducting in-depth experiments will enhance the credibility and significance of the findings. Furthermore, we extend an invitation to fellow researchers to join efforts in preparing and submitting this research to a reputable academic conference or journal. By collaborating and sharing insights,

we can collectively advance the field of point cloud classification and foster meaningful discussions within the scientific community.

References